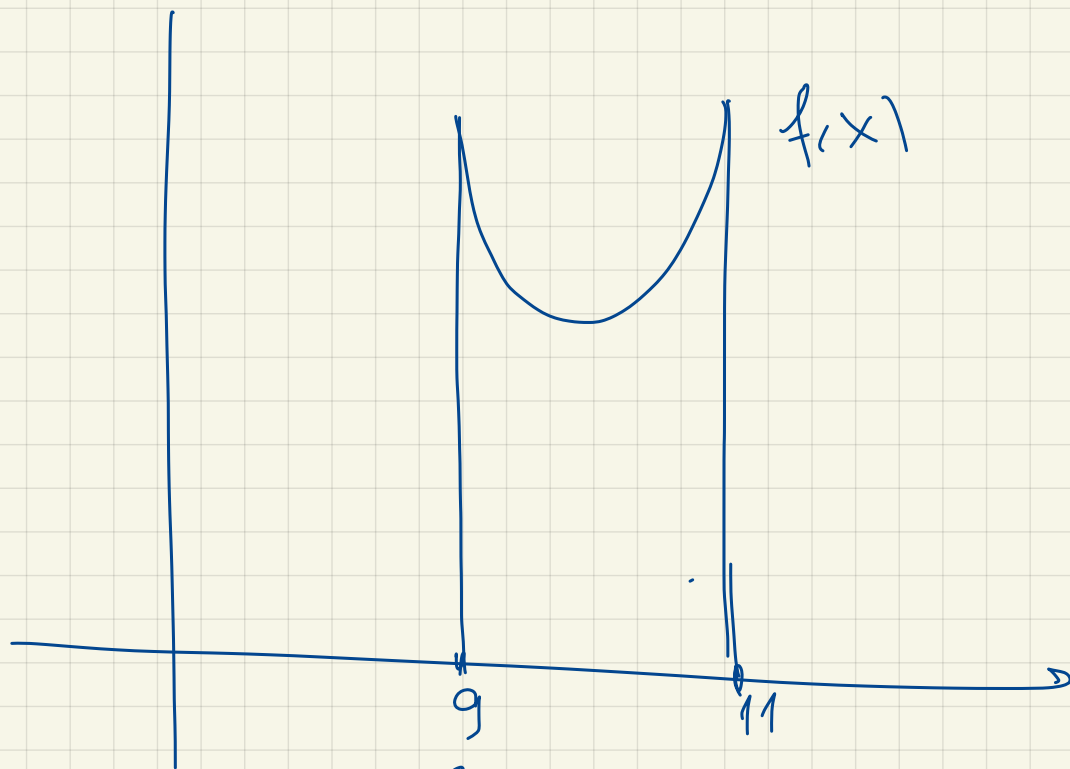






$$F(x) = x^3 - 30x^2 + 301x - \frac{1}{9}$$

$$\int_a^b f(x) dx = F(x) + C = \underline{F(b) - F(a)}$$

$$\int_9^{11} f(x) dx = F(11) - F(9) =$$

$$F(11) = 11^3 - 30 \cdot 11^2 + 301 \cdot 11 - \frac{1}{9}$$

$$F(9) = 9^3 - 30 \cdot 9^2 + 301 \cdot 9 - \frac{1}{9}$$

$$F(11) - F(9) = \underbrace{11^3 - 9^3}_{\cancel{30 \cdot 11^2 + 30 \cdot 9^2}} + \underbrace{301 \cdot 11 - 301 \cdot 9}_{\cancel{30 \cdot 11^2 + 30 \cdot 9^2}}$$

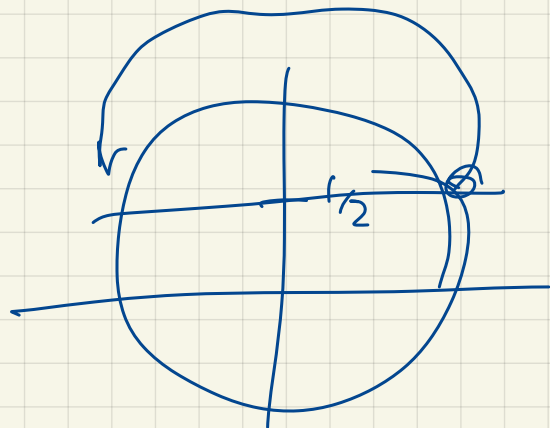
$$602 - 30(11^2 - 9^2)$$

$$602 - 1200 + 602 = \textcircled{4}$$

$$\textcircled{g} \quad v(t) = 7 \sin \frac{\pi t}{4} \quad \underline{\underline{>_{**} 3,5}}$$

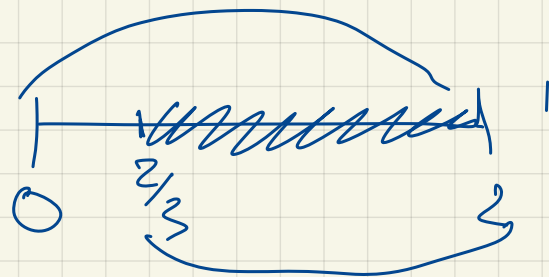
$$7 \sin \frac{\pi t}{4} > 3,5$$

$$\begin{cases} \sin \frac{\pi t}{4} > \frac{1}{2} \\ 0 < t \leq 2 \end{cases}$$



$$\frac{\pi}{6} < \frac{\pi t}{4} \leq \frac{5\pi}{6}$$

$$\frac{4\pi}{6\pi} < t < \frac{4 \cdot 5\pi}{6 \cdot \pi}$$



$$\frac{2}{3} < t < \frac{10}{3} > 2$$

$$\textcircled{\frac{2}{3} < t \leq 2}$$

$$\textcircled{2 - \frac{2}{3}}$$

$$\frac{2 - \frac{2}{3}}{2} = \frac{\frac{4}{3}}{2} = \frac{2}{3} = \textcircled{0,67}$$

11

$$\frac{7}{4}x - \frac{5}{4} = -13,5$$

$$\frac{7}{4}x = \frac{5}{4} - \frac{135}{10} =$$

$$\frac{7}{4}x = \frac{50 - 4 \cdot 135}{40}$$

$$x = \frac{50 - 4 \cdot 135}{70} = \frac{-490}{70} = \textcircled{-7}$$

$$\frac{\cos 3x - \sin x}{2 \sin x + \sqrt{2}} = 0$$

$$\underline{\cos 3x = \sin x}$$

$$\cos \alpha - \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cdot \sin \left(\frac{\beta - \alpha}{2} \right)$$

$$\cos 3x - \cos \left(\frac{\pi}{2} - x \right) = 0$$

$$2 \cos \left(\frac{3x + \frac{\pi}{2} - x}{2} \right) \sin \left(\frac{\frac{\pi}{2} - x - 3x}{2} \right) = 0$$

$$\begin{cases} \cos \frac{2x - \frac{\pi}{2}}{2} = 0 \\ \sin \left(\frac{\frac{\pi}{2} - 4x}{2} \right) = 0 \end{cases}$$

$$\begin{cases} x - \frac{\pi}{4} = \frac{\pi}{2} + \pi k \\ \frac{\frac{\pi}{2} - 4x}{2} = \pi k \end{cases}$$

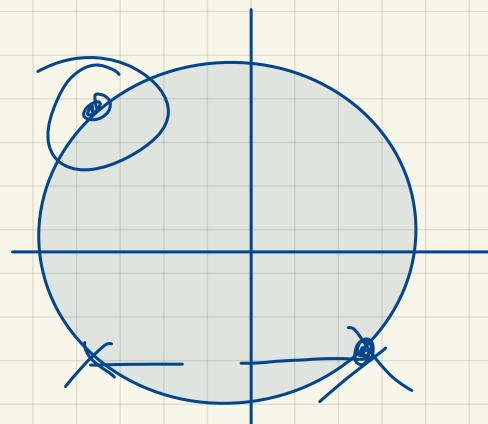
$$\begin{cases} x = \frac{\pi}{2} + \frac{\pi}{4} + \pi k \\ 2x = \frac{\pi}{2} - \pi k \\ x = \frac{\pi}{4} - \frac{\pi k}{2} \end{cases}$$

$$\begin{cases} x = \frac{3\pi}{4} + \pi k \\ x = \frac{\pi}{4} - \frac{\pi k}{2} \end{cases}$$

$$\sin x = -\frac{\sqrt{2}}{2}$$

$$x \neq \frac{5\pi}{4} + 2\pi k$$

$$x \neq \frac{3\pi}{4} + 2\pi k$$



$$\begin{cases} X = \frac{2\pi}{4} + 2\pi k \\ X = \frac{\pi}{8} - \frac{\pi}{2} \end{cases}$$

$$n = 8n.$$

18%
Donar

$$5.18$$

$$7\frac{5}{8} \cdot 1.18$$

$$1.18 \cdot 5\frac{5}{8}$$

$$1.14 \cdot 4\frac{5}{8}$$

14%
Bounn

$$0.185 + \frac{5}{8}$$

$$0.15 \cdot 7\frac{5}{8} + \frac{1}{8}$$

$$0.18 \cdot 5\frac{5}{8} + \frac{5}{8}$$

$$0.14 \cdot 4\frac{5}{8} + \frac{1}{8}$$

Donor

$$7\frac{5}{8}$$

$$5\frac{5}{8}$$

$$1.18$$

$$1.14$$

$$0.18$$

$$0.15$$

$$0.18$$

$$\Sigma = 2200$$

$$\sqrt{(y+ax-1)(y+ax)}$$

$$(y+ax-1)(y+ax) \geq 0$$

$$y = -ax + 1$$

(0:1)

$$y = -ax$$

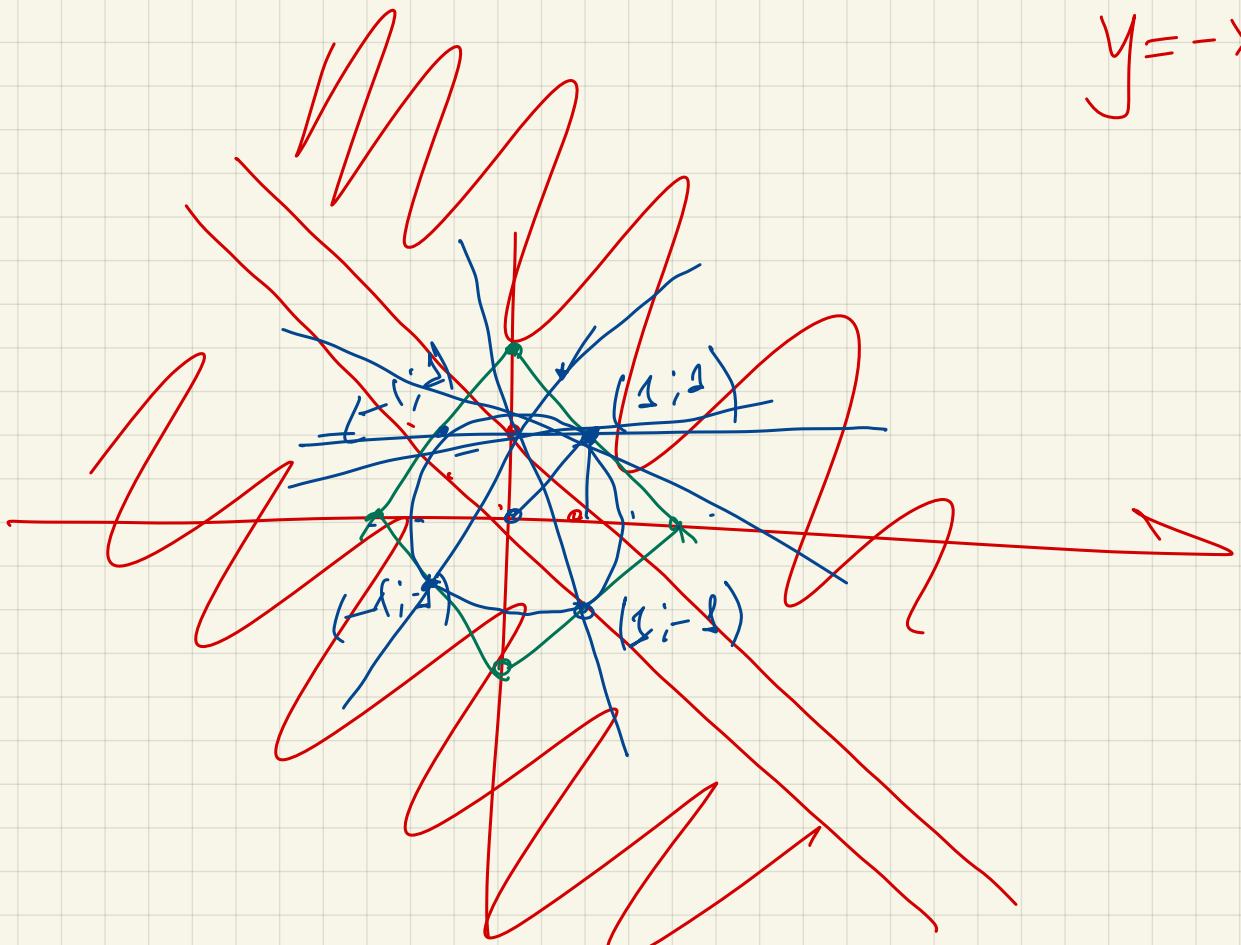
$\left. \begin{matrix} x=0 \\ y=0 \end{matrix} \right\}$

$$y \begin{matrix} +ax \\ -ax \end{matrix} = -ax + 1$$

$$a = 1$$

$$y = -x$$

$$y = -x + 1$$



$$y = -ax + 1$$

$$1 = -a + 1 \quad (1; 1)$$

$$a = 0$$

$$\begin{aligned} (-1; -1) \\ -1 = a + 1 \\ a = -2 \end{aligned}$$

$$\begin{aligned} -1 = -a + 1 \quad (1; -1) \\ a = 2 \end{aligned}$$

$$\begin{aligned} (-1; 1) \\ 1 = a + 1 \\ a = 0 \end{aligned}$$

$$2 \leq y \leq 2 + \sqrt{6x - x^2 - 5}$$

$$\sqrt{(x-1)^2 + (y-a)^2} + \sqrt{(x-5)^2 + (y-a)^2} = 4$$

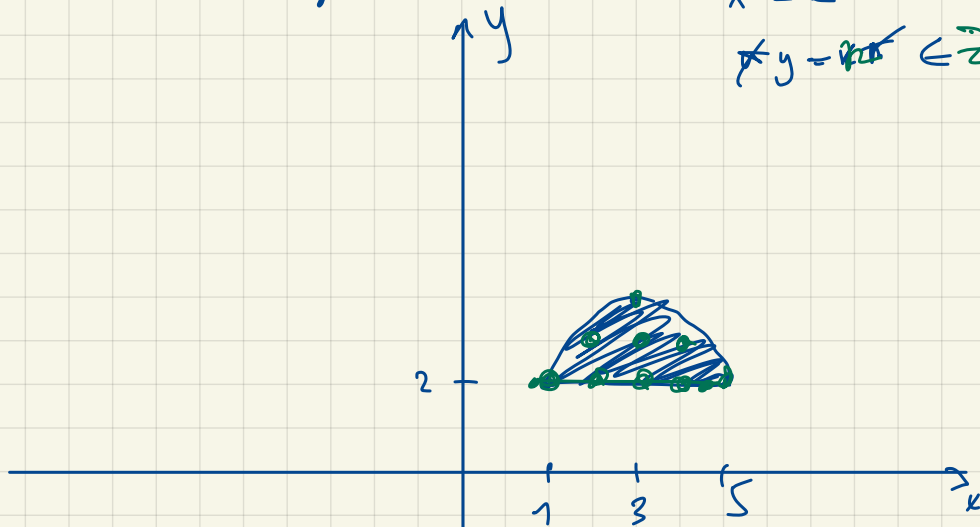
$$\sin x = 0$$

$$\cos y = 0$$

$$\pi x = \pi k$$

$$x = k \in \mathbb{Z}$$

$$y = \pi \in \mathbb{Z}$$



$$\begin{matrix} 2 \\ (1; a) \end{matrix}$$

$$\begin{matrix} 2 \\ (5; a) \end{matrix}$$

$$\sqrt{(1-5)^2 + (a-a)^2} = \sqrt{16} = 4$$

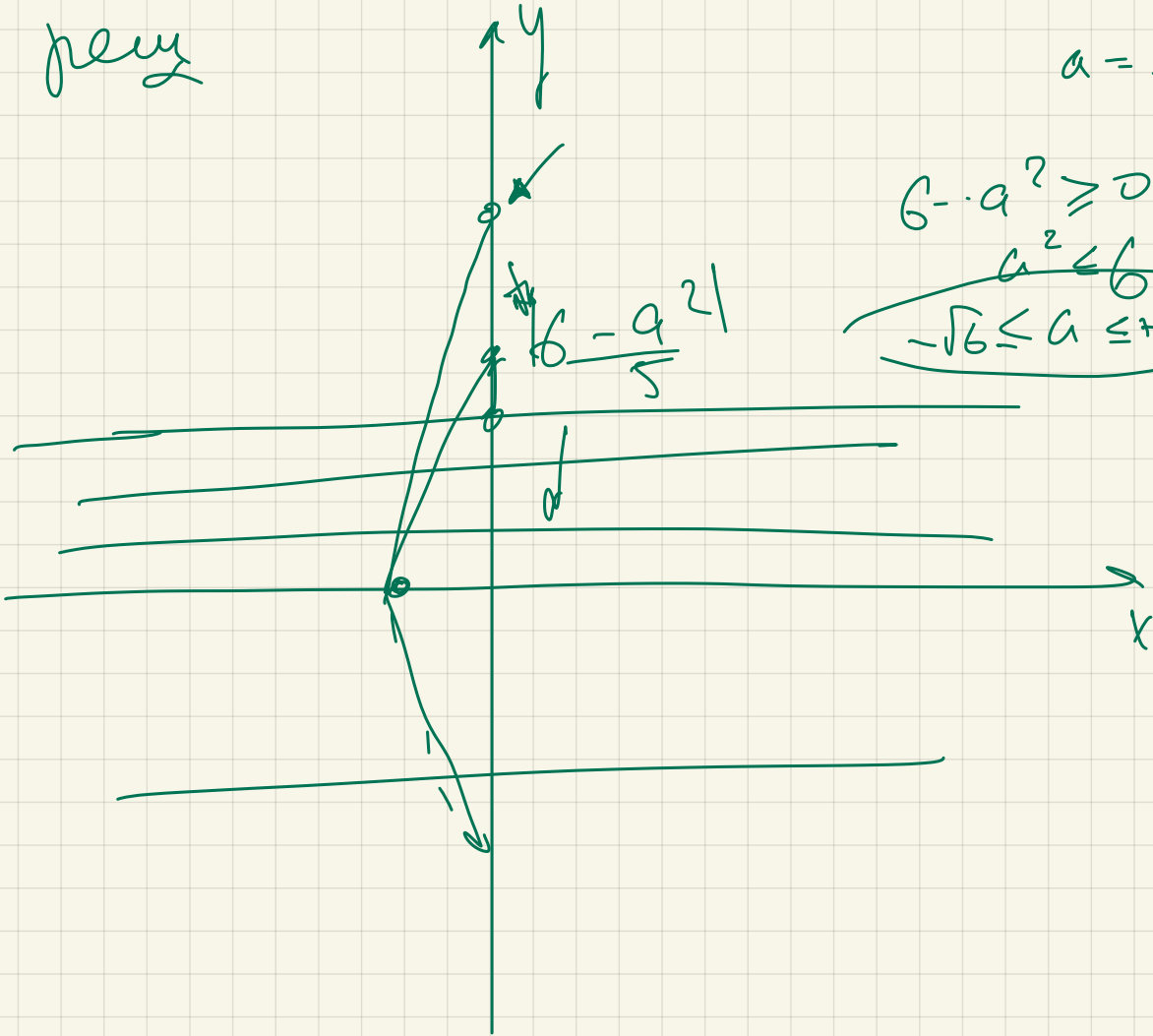
$$a = 2$$

$$\sqrt{(x+2)^2 + y^2} + \sqrt{x^2 + (y-a)^2} = \sqrt{4+a^2}$$

$$5y = |6-a^2|$$

$$y = \frac{|6-a^2|}{5}$$

1 peny



$$6 - a^2 \geq 0$$

$$a^2 \leq 6$$

$$-\sqrt{6} \leq a \leq +\sqrt{6}$$

$$(-2; 0) \quad (0; a)$$

$$a > \sqrt{6}$$

$$y = \frac{a^2 - 6}{5}$$

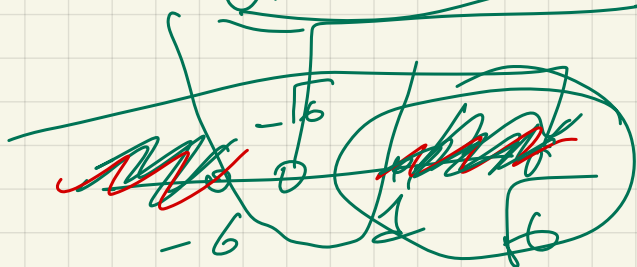
$$a > \sqrt{6}$$

$$a < -\sqrt{6}$$

$$0 \leq \left| \frac{6-a^2}{5} \right| \leq a$$

$$|6-a^2| \leq 5a$$

$$a^2 \leq 6$$



$$6-a^2 \leq 5a$$

$$a^2 - 5a - 6 \geq 0$$

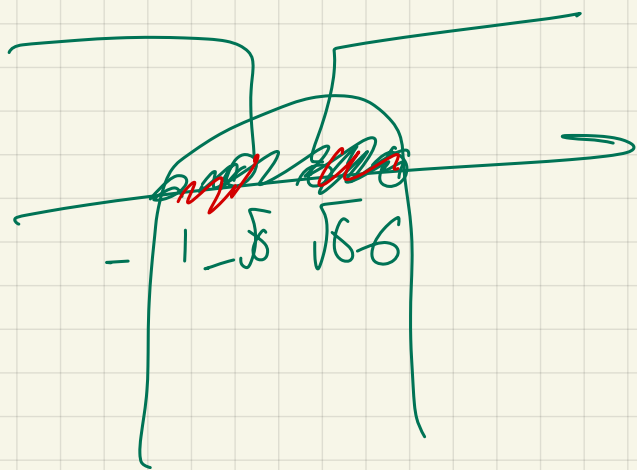
$$(a-6)(a+1) \geq 0$$

$$a^2 > 6$$

$$-6 + a^2 \leq 5a$$

$$-a^2 + 5a + 6 \geq 0$$

$$(a+1)(a-6) \geq 0$$



$$a \in [-1; \sqrt{6}]$$

$$a \in [-1; -\sqrt{6}] \cup [\sqrt{6}; 6]$$

$$a \in [1; 6]$$